

**BRAC University**  
**Dept. of Computer Science and Engineering**  
**Spring 2025**  
**CSE331-03**  
**QUIZ - 1(A)**

**Time: 20 minutes**

**Marks: 20**

Name:

ID:

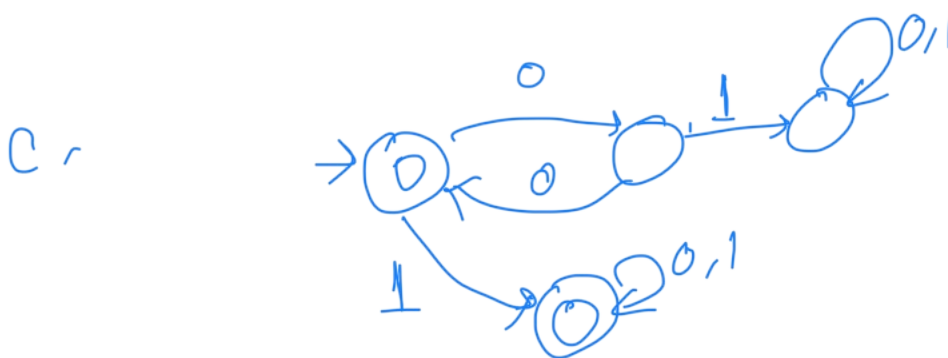
Section:

1. Construct DFAs that recognize the following languages.

- a.  $L = \{ w \in \{0,1\}^* : \text{the length of } w \text{ is two more than the multiple of } 3 \}$
- b.  $L = \{ w \in \{a,b\}^* : w \text{ contains "abb" as a substring} \}$
- c.  $L = \{ w \in \{0,1\}^* : w \text{ starts with even number of 0's} \}$

7

6



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**QUIZ - 1(B)**

**Time: 20 minutes**

**Marks: 20**

Name:

ID:

Section:

2. Construct DFAs that recognize the following languages.

- |                                                                                                      |   |
|------------------------------------------------------------------------------------------------------|---|
| a. $L = \{ w \in \{0,1\}^* : \text{the length of } w \text{ is one less than the multiple of } 3 \}$ | 7 |
| b. $L = \{ w \in \{a,b\}^* : w \text{ contains "baa" as a substring} \}$                             | 7 |
| c. $L = \{ w \in \{0,1\}^* : w \text{ starts with even number of } 1\text{'s} \}$                    | 6 |